MANAN BEDI

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EDUCATIONAL QUALIFICATIONS

Shaheed Sukhdev College of Business Studies, University of Delhi	Bachelor of Science with Honors, Computer Science Academic Excellence Award – 1st Position, GPA: 9.77/10	July 2022

TECHNICAL SKILLS

Languages: Python, R, Advanced SQL

Data Science Tech: Predictive Modelling, Statistical Analysis, Data Visualization (Power BI and Looker Studio), Machine Learning, Deep Learning, NLP, LLM, GenAI, MLOps (Grafana, Prometheus, MLFlow, Kubeflow, Docker), Prompt Engineering, Reinforcement Learning

Frameworks: Flask, Tensorflow, PyTorch,, TFOD

Technologies: Git/GitHub, Web Scraping, Prompt Engineering, AWS, Microsoft Excel, Google AppScript and AppSheet

WORK EXPERIENCE

- **Paytm [Team Lead – Senior Product Analyst (UPI P2P Product) – Full Time – Delhi, India]** **[May 2022 - Present]**
  - **Leading Product Ownership** for core UPI P2P journeys with **~80M+ Monthly Active Users** across Android, iOS – driving transaction stability , funnel conversion, and GMV growth.
  - **Architected** and scaled **funnel instrumentation** across **15M+** daily P2P events by building **Python** pipelines and **advanced SQL** queries to capture real-time **L1–L3 drop-offs**, retries, API latency distributions, and NPCI response codes.
  - **Redesigned and launched major UI/UX overhauls** across To Mobile, Bank, Self, and Chat Send flows, leading to a **+14.3% uplift in Send Money Initiation Rate**, **+17.1% MoM growth in P2P GMV**, and a **conversion funnel boost from 72.1% to 86.4% PPS rate**, outperforming industry benchmarks. **Coordination** with Engineering, Design and Product Teams for quick feature rollout.
  - Introduced **AI-driven Reminder Suggestions** and **Personalized Favorites**, lifting **repeat P2P retention by 15.8%** and reducing transaction journey re-entry time by 2.6s. Developing **Predictive Models** using **ML** to forecast future trends.
  - Launched **failover diagnostic tooling** for NPCI Mapper errors - auto-tagging top failing BINs, PSP handles, and banks. Reduced mapper-related transaction failures by **19.4%**.
  - **Performance Tracking:** Tracking key metrics like GMV, MTU, Transactions, Deployed Devices to **identify** trends and reporting
  - **Data Visualization:** Creating Dashboards on **Looker Studio** that present complex data in an interactive format for Investors.
  - **Automation:** Developed Python and Google AppScript solutions to automate analysis and reporting workflows, **reducing turnaround time from 2 days to 2 minutes with zero errors.**
  - **Cost Optimization:** Led AWS cost tracking and savings of ~₹1 Cr; built a custom Azure OpenAI LLM app to automate analytics.
- **Socio Global Brands Inc., Canada (Manager - Technology)** **[Mar 2021 – Oct 2021]**
  - Worked on a **Centralized Application** for sending and receiving donations for various Institutions based in **US and Canada**
  - Developed Backend using **Python Flask** Framework, integrated with PostgreSQL DB, and frontend using HTML, CSS, Bootstrap
  - Added features like **Stripe** PG Integration, Mass Emailing List, Personalized Pages, Encryption, etc.
- **Let the Data Confess Pvt. Ltd. (Data Science – Intern)** **[Sep 2021 – Jan 2022]**
  - Authored a series of engaging and informative blog posts on various Data Science topics, showcasing a grasp of complex concepts.
  - Achieved remarkable success on Quora by garnering over **20,000** views within just two months for solving queries.
- **AnalytixWare Technologies (Data Science – Intern)** **[Mar 2021 – Jun 2021]**
  - Worked on a topic-based document similarity project using NLP. Experimented with multiple document similarity models and techniques like Cosine-Similarity to identify the similar documents of documents provided leading to an organized collection.

PROJECTS (AND PUBLICATIONS)

- **Legal Text Summarizer (NLP)** **[Link]**
  - **Implemented a Legal Text Summarizer using NLP**, fine-tuned HuggingFace APIs to improve legal document comprehension, and evaluated output quality on Rough Score benchmarks.
  - **Developed** the MLOps pipeline with MLFlow, Docker, and Prometheus to package, track, and deploy the application efficiently.
- **Customer Churn Prediction (ML)** **[Link]**
  - Developed Churn Prediction system, optimizing customer retention efforts for business growth.
  - Integrated with Web using Flask Framework and deployed on Heroku. Achieved more than **90%** accuracy.
- **Flipkart Product Review Scraper** **[Link}**
  - Created a Flipkart Product Review Scraper using **BeautifulSoup** Library, enabling user-friendly product review extraction from DB.
  - Created Web Application for it using Flask and deployed on Heroku Cloud
- **Multi-Agent AI Sales Automation System (using n8n and ElevenLabs)**
  - **Architected a hierarchical multi-agent AI system** using n8n orchestration and LLM-based role specialization, enabling autonomous task delegation, tool invocation, and end-to-end workflow automation.
  - **Developed an automated lead intelligence pipeline** integrating Firecrawl web scraping, Hunter.io email verification, and Pipedrive REST APIs for structured CRM ingestion and synchronization.
  - **Built voice-enabled conversational agents** using ElevenLabs with Google Calendar API integration, applying prompt alignment and structured outputs for reliable lead qualification, demo scheduling, and CRM logging.
- **Self Driving Car Agent (Reinforcement Learning)** **[Link]**
  - Developed a self-driving car agent model using the OpenAI Gym Toolkit and trained it using the PPO (Proximal Policy Optimization) algorithm from the stable_baselines3 module
  - Trained the model over 1.5 million timesteps, achieving a high performance score of 680
- **YouTube Video Script Generator (LLM and GenAI)** **[Link]**
  - Developed a YouTube Video Script Generator using LangChain, enabling users to effortlessly create compelling video scripts by simply entering a topic, with prompt templates and Few-Shot Learning enhancing the quality and accuracy of generated scripts.
  - Integrated a flexible choice of models: OpenAI API (user-provided key) or Hugging Face API (default), powered by a Text-to-Text Generation Model; built an intuitive UI using Streamlit for a seamless user experience.

PUBLICATIONS

- **Application of Deep Learning Techniques for COVID-19 Management [Springer]** **[Link]**
  - Understanding COVID-19: The Role of Computational Intelligence Research Paper (Published in Springer). More than 50 citations
- **Analysis of COVID-19 Tracking Tool in India: Case Study of Aarogya-Setu Mobile Application [ACM-Journal]** **[Link]**
  - The paper analyzes AI Powered features incorporated into the Application for COVID-19 Tracking in India. 50+ Citations and 12k Downloads
- **YouTube Videos Clickbait Detection [Under Review]**
  - Conducting research on YouTube video clickbait detection using Deep Learning and Large Language Models (LLMs) by implementing techniques such as video chunking and optimizing hyperparameters to improve model performance.
  - Fine-tuning parameters such as learning rate, batch size, and sequence length, while leveraging chunked video data for efficient processing, to develop a scalable, high-accuracy solution for automated content classification and clickbait identification.
- **Spatiotemporal GCNs for VANETs (Vehicular Ad Hoc Networks) [Under Review]** **[Link]**
  - Contributed to a research project focused on modeling spatiotemporal traffic patterns in VANETs using **GCN + GRU-based architectures**.
  - **Designed and implemented a novel SpatioSpectral Attention with GCN model** to enhance dynamic traffic flow prediction and vehicle-to-vehicle communication accuracy.
  - Utilized **PyTorch Geometric** and attention mechanisms to improve forecasting of link reliability and traffic congestion.
  - Worked with over **1500 time steps** of simulated mobility data, enabling reliable spatiotemporal learning and edge classification in dynamic networks.
  - Work contributed as part of **ScholarsColab's collaborative research initiative**.
- **Sentiment Analysis: Lexicon to Generative AI Approaches**
  - Co-contributed to a chapter providing a comprehensive review of sentiment analysis techniques ranging from lexicon-based methods to transformer-based LLMs like GPT, BERT, and RoBERTa.
  - Covered use cases across healthcare, politics, finance, education, and cybersecurity, with emphasis on the challenges and opportunities of applying GenAI in sentiment modeling.

SPEAKER SESSIONS

- Developers Week Conference *(Santa Clara, California, United States of America) [Selected one of the 30 out of 500+ submissions]* *[Link]*
- *TDWI Conference (Munich, Germany) [Selected one of the 50 out of 700+ submissions]* *[Link]*
- React Delhi Conference (Microsoft Office, Gurugram, India)
- Association for Computing Machinery Chapter (University of Delhi)
- GDG SRMIST (SRM Institute of Technology)
- GDSC SRCSAW (University of Delhi)

AWARDS AND ACHIEVEMENTS

- **1st position -** Class 12th (Science Stream) – Lovely Public Sr. Sec. School, P.D. Vihar, New Delhi
- **Most Innovative Project –** Hackathon (Out of 45+ teams) – SGGSCC, University of Delhi
- **Rising Star of the Month (Paytm) –** Award given for automating various reports to reduce the overall Turnaround Time and taking ownership
- **1st Position –** Bachelors in Computer Science [University of Delhi] and got a Merit-based renumeration amount by Government of Delhi
- **ML Contributor –** One of the 50 contributors (Leveraging ML to predict accomplishment rate of startup) at Omdena Pennsylvania, USA
- **Judge and Mentor –** Judged and mentored multiple hackathons around Globe including Hack the North, Canada and NASA International Space Apps Hackathon. Also mentored various Open-Source projects
- **President (ACM-SSCBS) -** Co-founded the SSCBS Student Chapter. Led a team of 12 people interested in research
- **Meeting with Former President of India –** Got an opportunity to meet the Former President of India Late Dr. Pranab Mukherjee on the basis of merit

Volunteering Experience

- **Learner Success Head (Linux World India] –** Helped resolve queries of students to excel in the program of AI offered by Linux World India
- **Python Specialist Tech Volunteer [Linux World India] –** Helped resolved technical queries of the students
- Volunteered to get the medicines delivered to the COVID-19 Patients during the Pandemic . Also maintained a database of the medicines and Oxygen Cylinders available